Start with the primal constraints

$x_1 + x_2 \le 4$ (flour) $\qquad 2x_1 + x_2 \le 5$ (sugar)

multiply by $y_1, y_2 \ge 0$ and add

Combine them into one inequality

$$(y_1 + 2y_2)\, x_1 + (y_1 + y_2)\, x_2 \quad \le \quad 4y_1 + 5y_2$$

choose y so coefficients cover 3 and 2

Make the left side dominate the profit

$$y_1 + 2y_2 \ \ge\ 3 \qquad y_1 + y_2 \ \ge\ 2$$

$$\Longrightarrow \quad 3x_1 + 2x_2 \quad \le \quad 4y_1 + 5y_2 \quad \text{(a certificate!)}$$

minimize the bound over all valid y

Hunt for the best certificate

$$\min\ 4y_1 + 5y_2 \quad \text{s.t.} \quad y_1 + 2y_2 \ge 3, \quad y_1 + y_2 \ge 2, \quad y \ge 0$$

This is the dual LP!